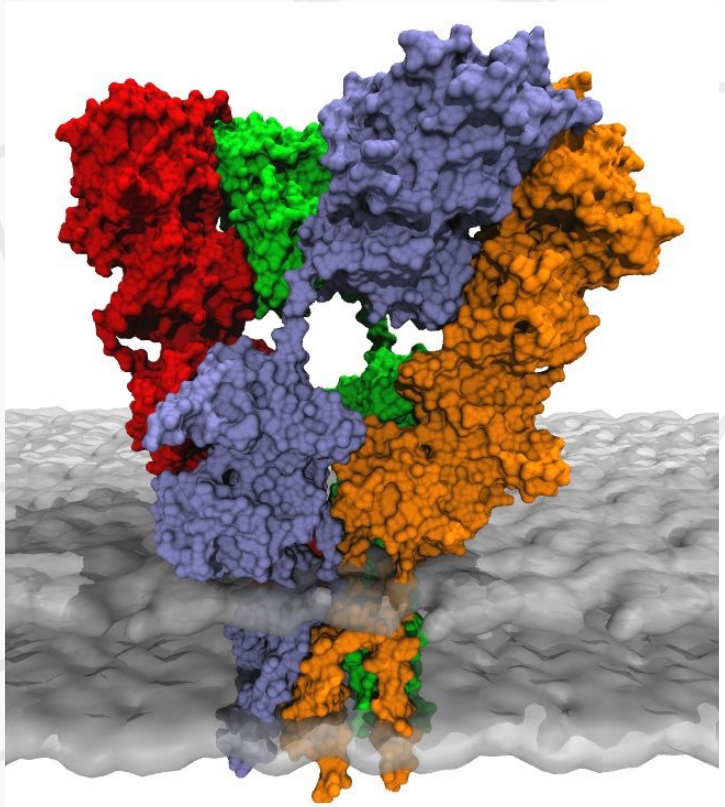
Ligand-protein binding

David Gloriam
Professor, PhD
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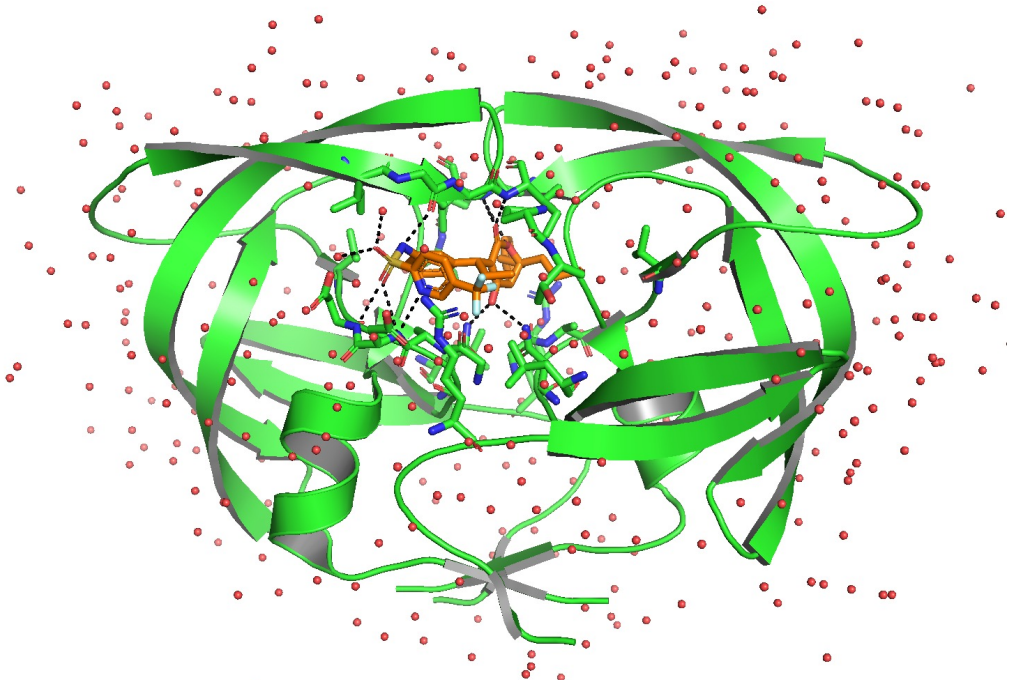
UNIVERSITY OF COPENHAGEN



Lecture learning goals

Background for:
- **Exercises**
- **Classes**

- What are the molecular interactions of ligands and proteins?
- Which forces contributions to the binding free energy (ΔG)?



Binding of tipranavir to the protein (HIV-1 protease)

Video



https://www.youtube.com/watch?v=dyxoQWknNJ8

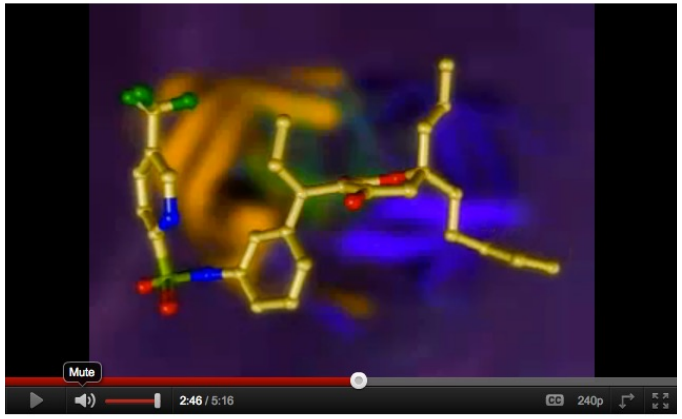
Question

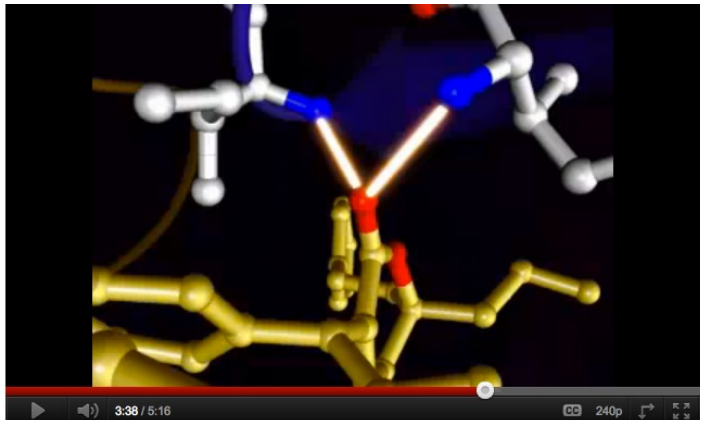
- Why does the ligand tipranavir bind to the protein HIV-1 protease?

Why does the ligand tipranavir bind to the protein HIV-1 protease?

"Unbound Tipranavir displays the **bioactive conformation** more often than older HIV-protease inhibitors"

Tipranavir makes a direct interaction with the Ile backbone atoms **replacing a water molecule**..





Binding events

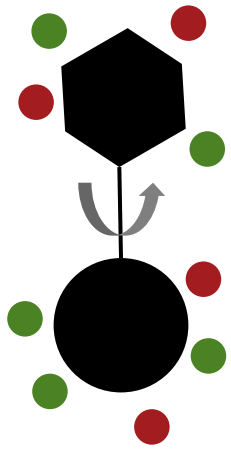
Unbound

Ligand in solution

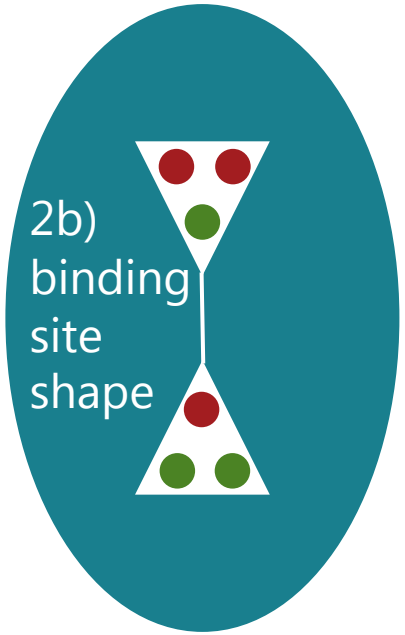
1a) Stable water

1b) Loose water

2a) Rotatable bond

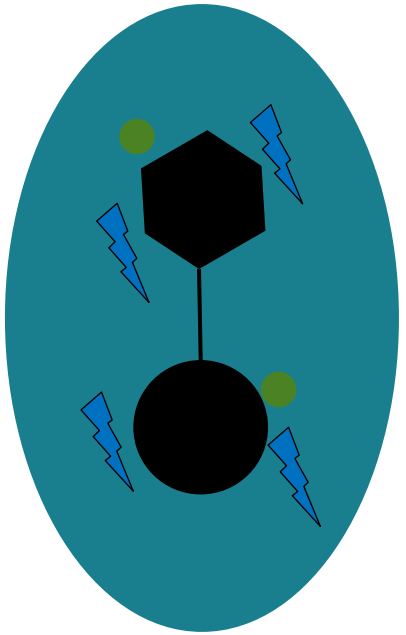


Protein in solution



Bound

Ligand-protein complex



3) Molecular interactions

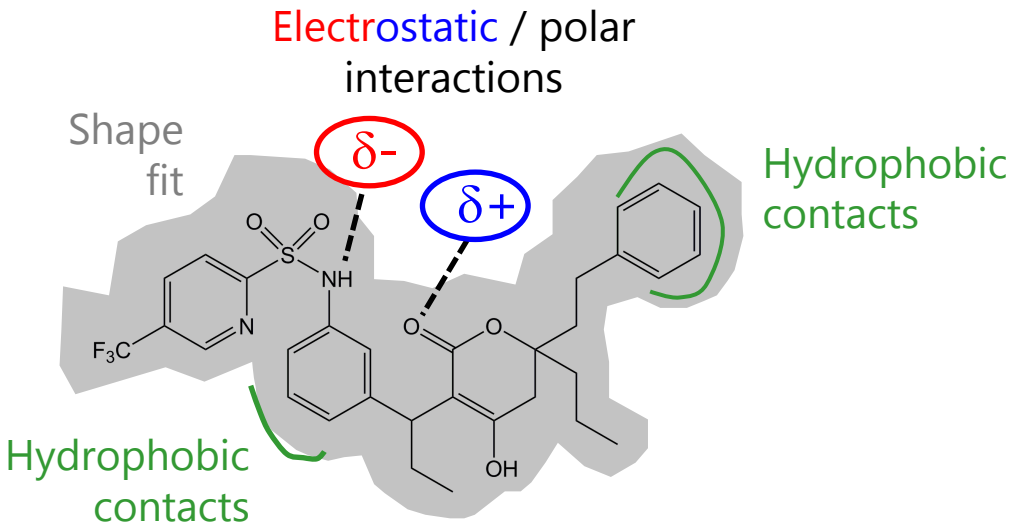
Events

1. Desolvation
Loose waters: Removal gives energy
Stable waters: Removal costs energy

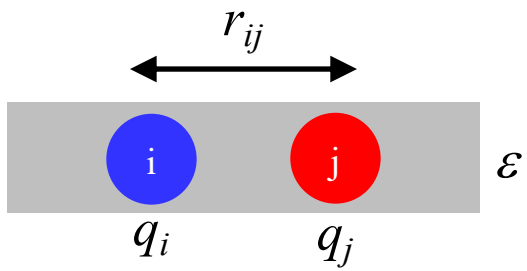
2. Conformational change
Ligand: adopts bioactive conformation
Protein: induced fit of binding pocket
Both cost energy

3. Molecular interactions
Mostly favorable – gives energy
Mix of polar or hydrophobic

Ligand-protein interactions



Electrostatic / polar interactions (ΔG_{polar})



q_i: Charge (on atom/ion i)

q_j: Charge (on atom/ion j)

r_{ij}: Distance between two charges

e: Dielectric constant

78 in water

1 in gas phase

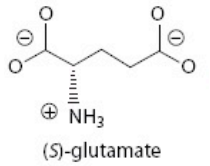
Calculating the energy

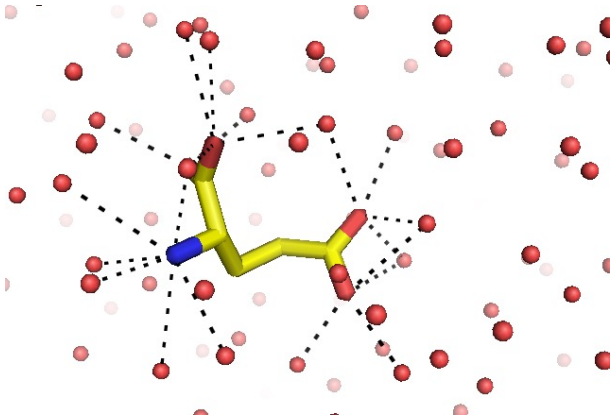
Coulomb's law

$$E_{polar} = \frac{q_i q_j}{\varepsilon r_{ij}}$$

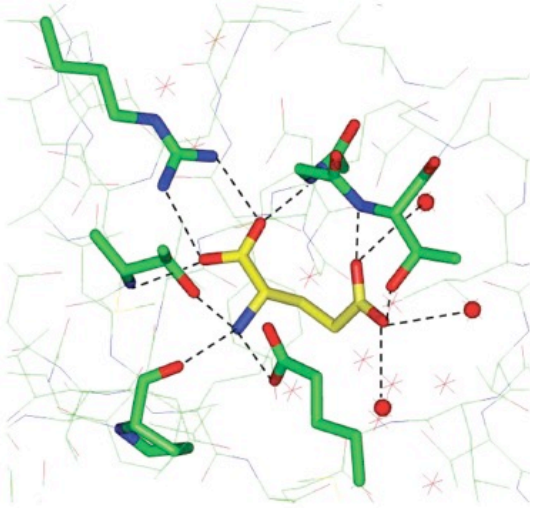
Water and protein compete for ligand hydrogen bonds

Ligand in water





Ligand in protein binding site

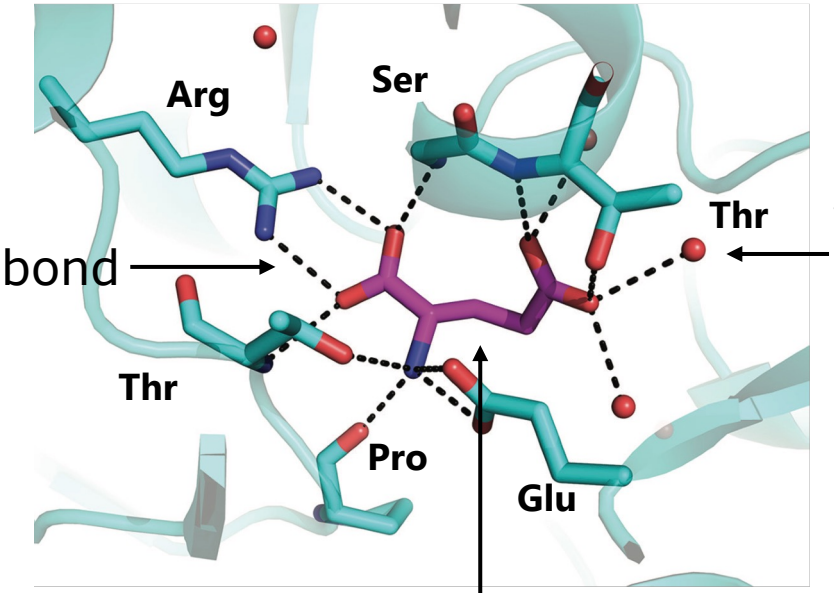


A polar ligand prefers water, and has a negative ΔE when binding the protein

A hydrophobic ligand avoids water, and has a positive ΔE when binding the protein

Hydrogen bond – and electrostatic interaction

(S)-glutamate bound in the AMPA receptor GluA2

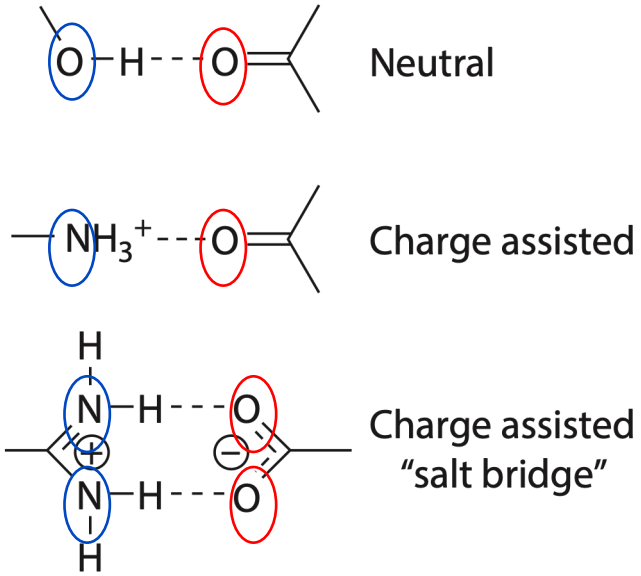


Electronegative atoms (e.g. O/N)

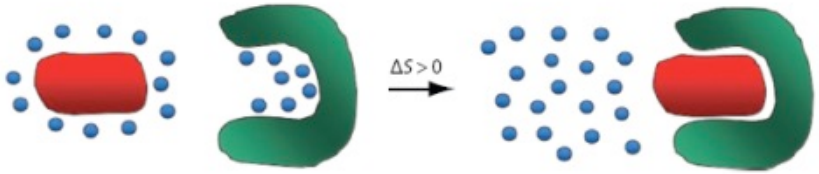
X–H --- Y

Hydrogen 2.5–3.5 Å distance

Hydrogen bonds can be combined with charge



Hydrophobic effect ($\Delta G_{hydrophobic}$)

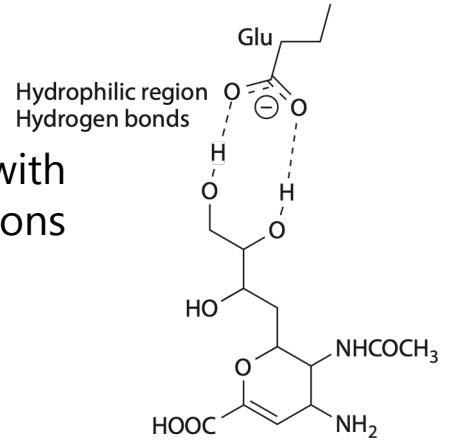


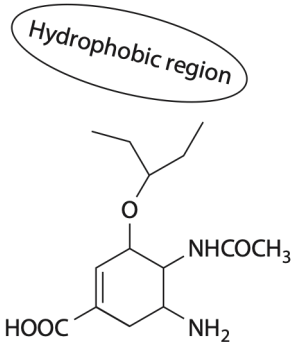
Hydrophobic ligands and protein binding pockets get a favorable change in entropy (ΔS) when binding.

This increases binding affinity

IC_{50} = 150 nM (worse)	IC_{50} = 1 nM (better)

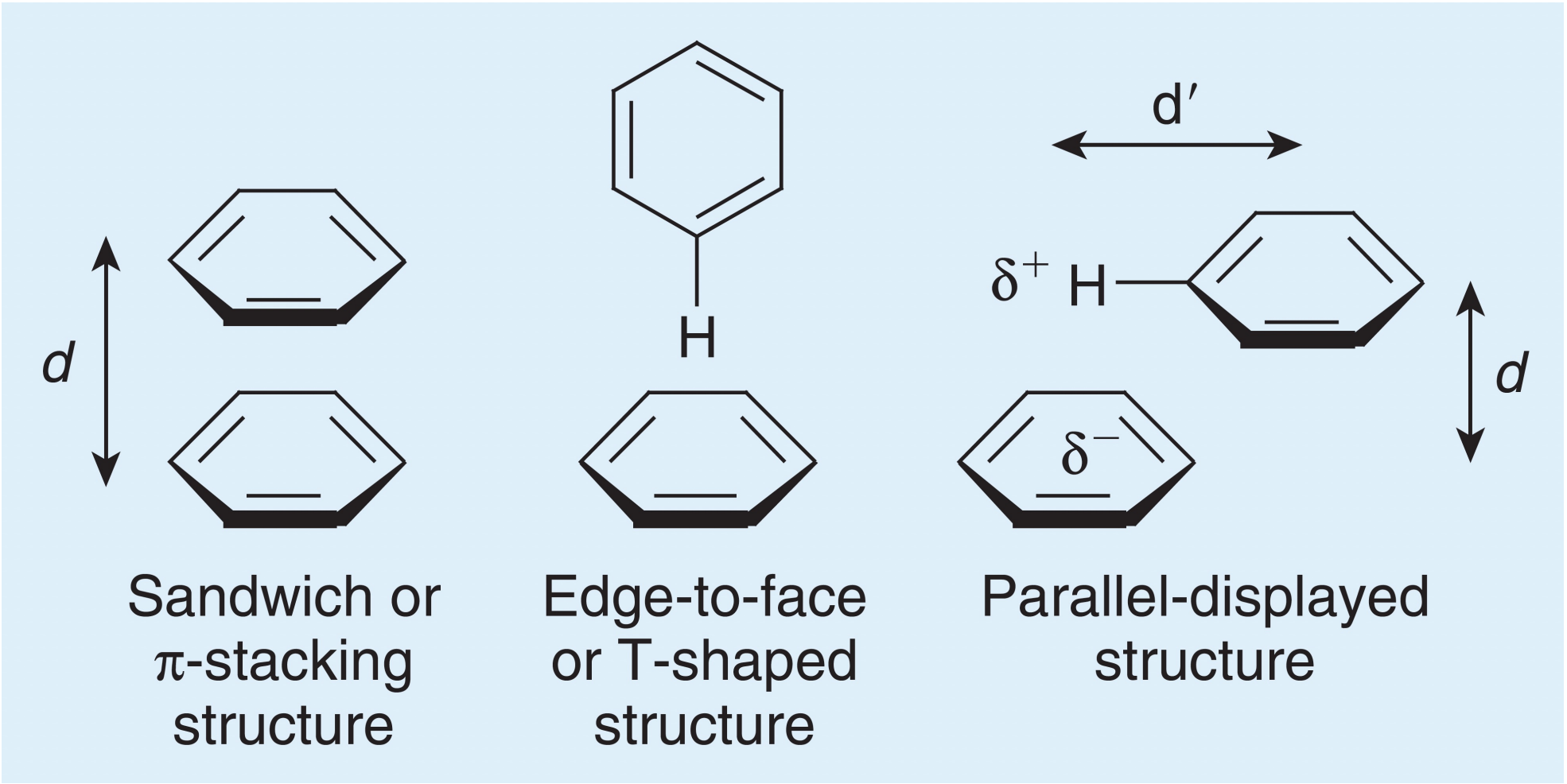
Ligand with polar interactions



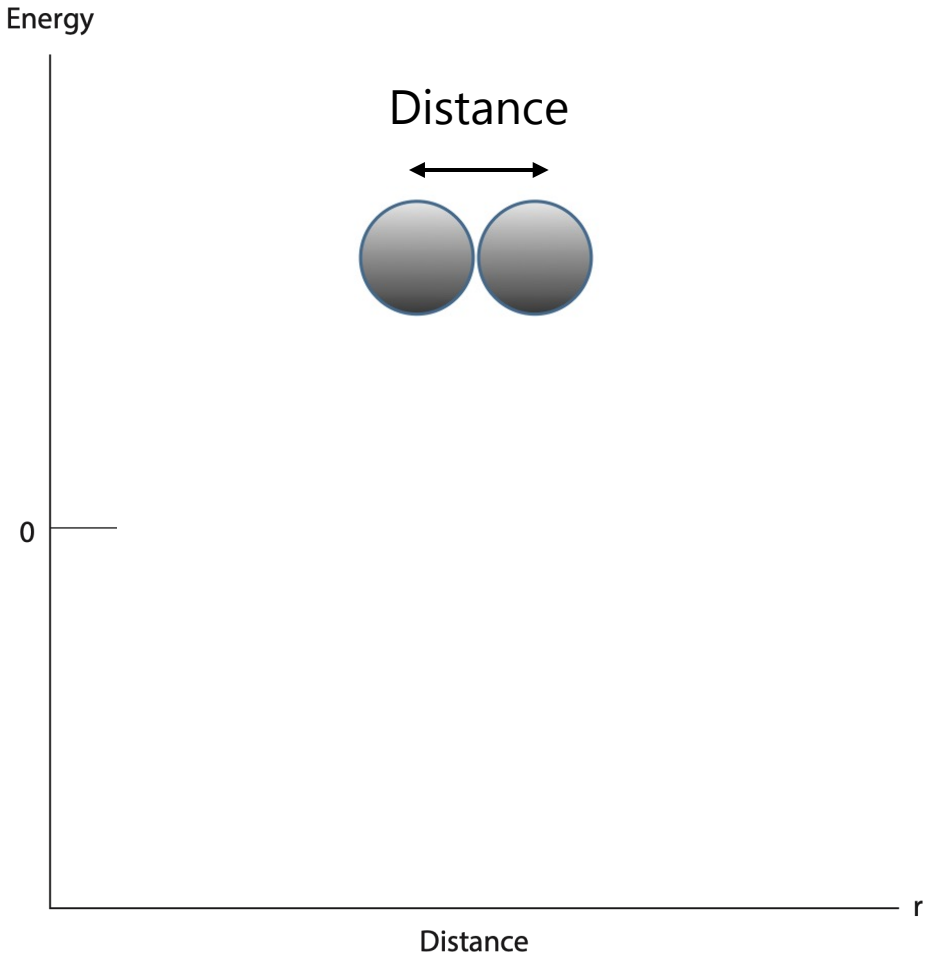


Ligand with hydrophobic interactions

Hydrophobic contacts of aromatic rings

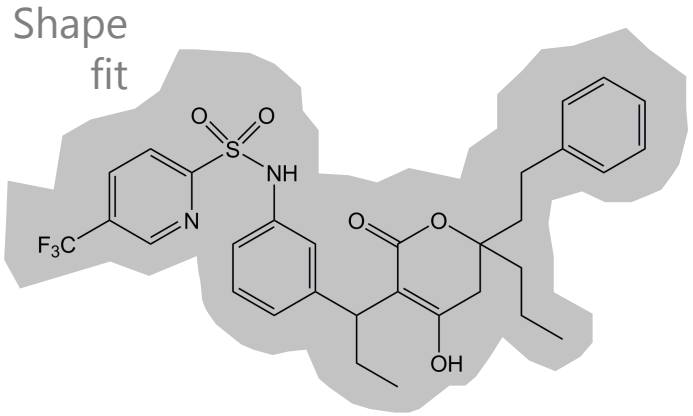


Shape complementarity via van der Waals contacts (ΔGvdW)

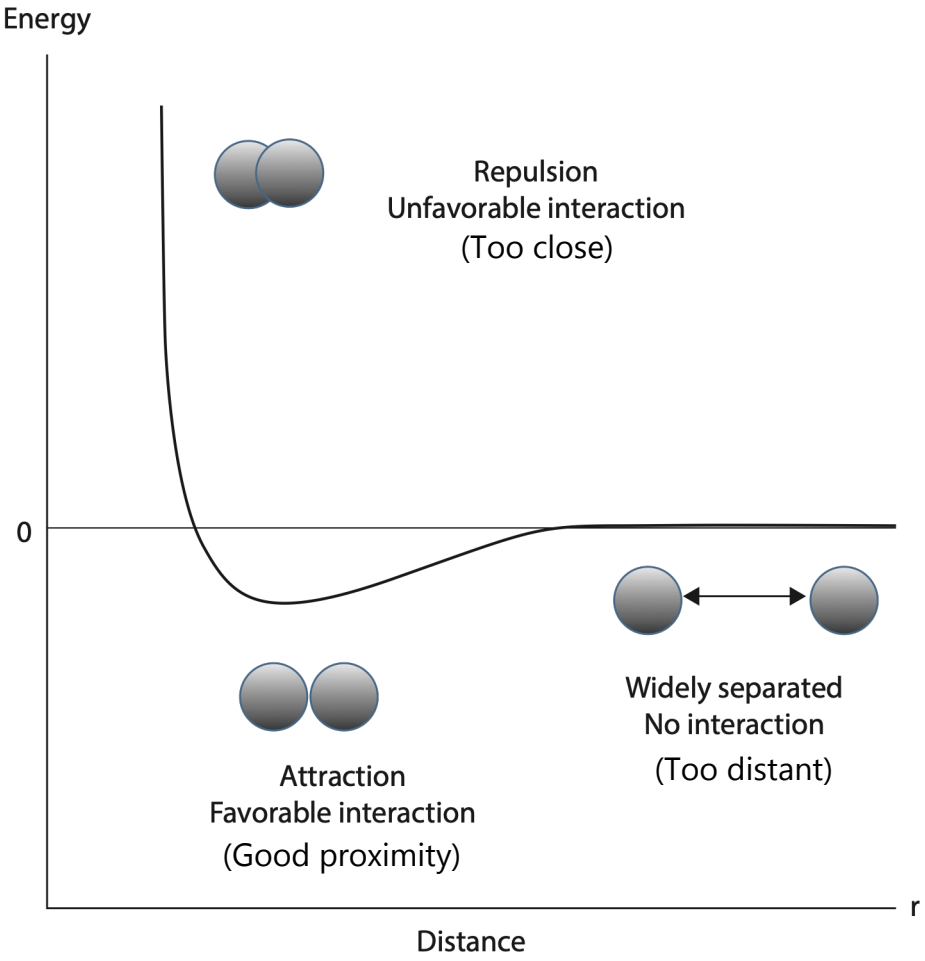


Question

How do you think that energy of two atoms increases or decreases based on distance?



Shape complementarity via van der Waals contacts (ΔGvdW)



Repulsive term, short distances

Attractive term ~ 4Å for C–C interactions

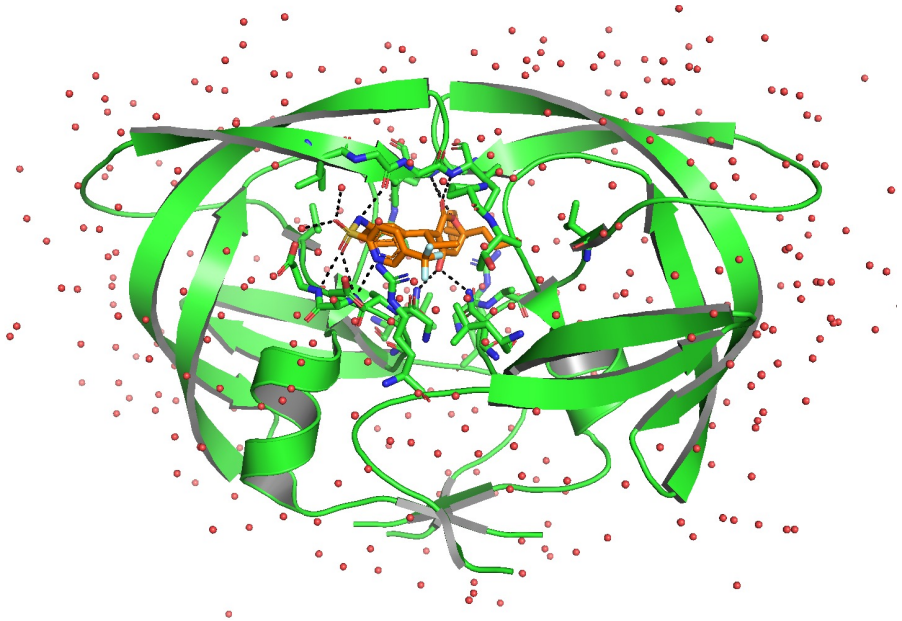
$$E_{vdW} = \frac{A}{r^{12}} - \frac{B}{r^{6}}$$

- Typical strengths for atom-atom interactions: 0.2 kJ/mol - many good contacts will contribute to affinity

Summary
Binding affinity, ΔG is the sum of all contributions

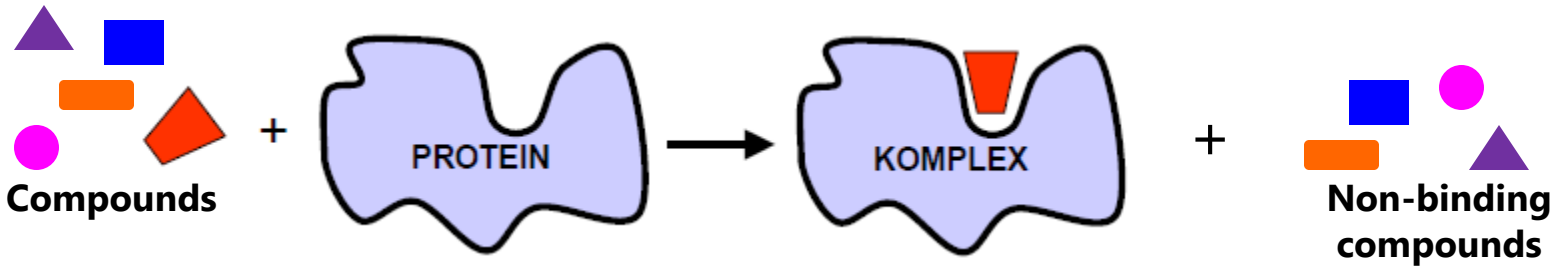
Penalty for freezing overall motion | Conformational penalties | Electrostatic interactions | Hydrophobic effects | Shape complementarity

$$\Delta G = \Delta G_{translational+rotational} + \Delta G_{conformation} + \Delta G_{polar} + \Delta G_{hydrophobic} + \Delta G_{vdW}$$



Question on docking (of ligands into proteins)

Docking simulates binding of ligand into a protein 3D structure



The protein is kept rigid whereas compounds are docked flexibly.

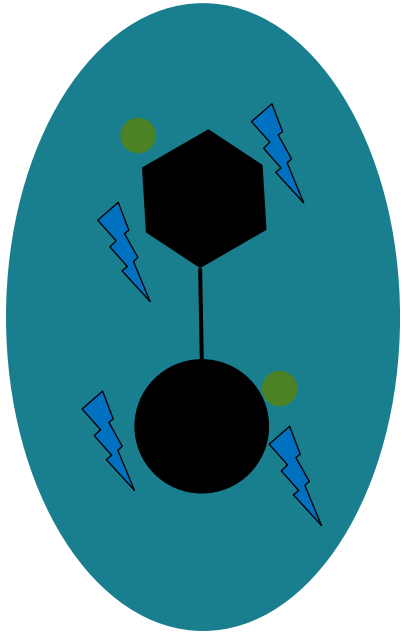
Water molecules are not part of docking.

Question

Which contributions to binding affinity does docking exclude?

Force fields
Estimate ligand-protein interaction energy

Ligand-protein complex

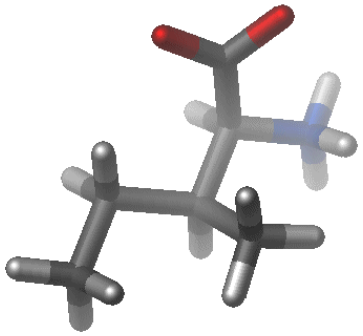


3) Molecular interactions

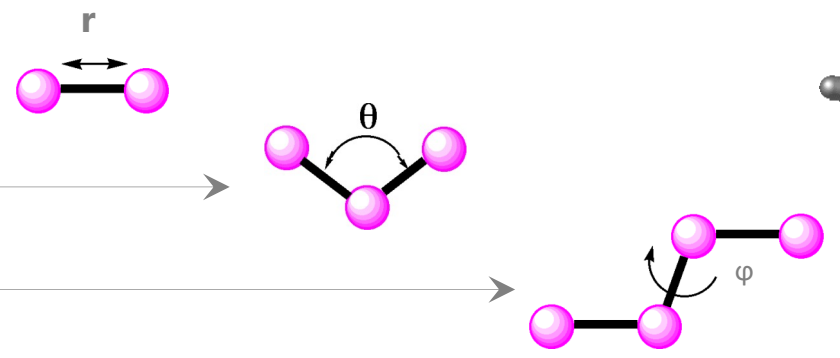
- Function and parameter sets used to calculate the potential energy of a system of atoms in molecular mechanics and molecular dynamics simulations.
- The parameters of the energy functions may be derived from experiments in physics or chemistry, calculations in quantum mechanics, or both.

Basically a set of equations and constants

Force Field terms



- Bonded terms
  - Bond lengths
  - Bond angles
  - Torsional angles
- Non-bonded terms
  - van der Waals
  - Electrostatics





- Conformational energy = Sum of bonded and non-bonded energies

This question concerns ligand-protein binding. Which of the TWO statements below are TRUE?

a) Increasing the hydrophobicity of a ligand can potentially increase its binding affinity to the primary protein target.

b) Increasing the hydrophobicity of a ligand typically increases its selectivity for the primary protein target.

c) Upon ligand-protein binding, the ligand but not protein changes conformation.

d) Strong electrostatic interactions, hydrophobic effects and shape complementarity typically decrease binding affinity.

e) Freezing of motion and large conformational changes of the ligand typically increase binding affinity.

f) If a ligand has many hydrogen bonds to water molecules, when unbound, but few hydrogen bonds when bound to the protein target this typically results in an energy penalty.

This question concerns ligand-protein binding. Which of the TWO statements below are TRUE?

a) Increasing the hydrophobicity of a ligand can potentially increase its binding affinity to the primary protein target.

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Teaser for my next lecture – on structure-based drug design

Methods covered

AlphaFold
RoseTTAFold

Site maps

Docking

Virtual screening

Molecular dynamics

Videos

Binding of tipranavir to HIV-1 protease

https://www.youtube.com/watch?v=dyxoQWknNJ8

Book sections

Book

The Handbook of Medicinal Chemistry: Principles and Practice.

Ward, Simon E; Davis, Andrew (Eds.)

2th Edition, 2023, The Royal Society of Chemistry England

Sections

3.4 Modeling Solvation

3.5 Conformations, Conformational Energy and Drug Design

3.7 Docking and scoring functions

4.4 Theory and Practice of Structure-based Design

Old exam question

Add

- Shape-fit, important for selectivity

Next year

- Add slide on conformational restriction of ligands
- Tipranavir-HIV-1 video is not very informative, and old – replace